FACET: Measuring Attribution Faithfulness in Multi-Factor LLM Reasoning

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Abstract

Many decisions require weighing $N \geq 5$ different factors simultaneously. When evaluating whether LLMs integrate all factors or collapse onto a single canonical one, a common measurement is to ask the model "which single factor is most important?" We show this probe format produces spurious attribution artifacts that misrepresent the underlying model behavior. We introduce **FACET** (Factor Accessibility, Coverage, and Evidence Test), a benchmark that compares two probes on the same cases and models: a single-factor probe and a weighted-rank probe that asks the model to assign percentage weights to all N factors. Evaluating nine frontier configurations from six labs (Anthropic, OpenAI, Meta, Mistral, DeepSeek, Alibaba/Qwen) on ten real legal balancing cases and three counterfactual variants, the single-factor probe produces apparent lexicographic collapse: models lock onto a textbook-canonical factor and 0/8 shift their attribution when that factor is explicitly neutralized on a Barker-style risk-utility counterfactual. The weighted-rank probe, applied to the same models and cases, shows the opposite: **8/8 assign near-zero weight to neutralized factors on all three counterfactuals** and produce distributed weights (20–30% for the top factor, not 100%) on real cases, with no single factor receiving $\geq$50% of the weight from any model. The "canonicity-dependent dissociation" visible under the single-factor probe is a probe-format artifact, not a model property. We argue that prior findings of attribution unfaithfulness in multi-factor reasoning should be re-examined under weighted-rank probing.

1 Introduction

1.1 The Problem

Many decisions require weighing $N \geq 5$ qualitatively different factors against one another: a physician balancing symptoms, lab results, and patient history; a compliance officer weighing risk indicators across regulatory dimensions; a judge integrating doctrinal factors in a balancing test. When LLMs are deployed in these settings, a core question arises: *does the model actually integrate all N factors, or does it collapse onto whichever single factor is most canonical in its training data?*

Both strategies can produce the same correct answer on typical cases, so outcome accuracy alone cannot distinguish them. The key methodological challenge is constructing cases where the two strategies predict different outcomes.

1.2 Why Legal Balancing Tests

We instantiate FACET in the domain of US tort law because legal balancing tests offer three properties that make them an ideal testbed for multi-factor reasoning:

1. **Explicit factor lists.** Doctrines like *Rowland v. Christian* (1968, seven factors for duty of care), *Barker v. Lull Engineering* (1978, five factors for design defect), and *Biakanja v. Irving* (1958, six factors for third-party duty) enumerate their factors by name. This eliminates ambiguity about what the model should be weighing.
2. **Verifiable ground truth.** Appellate court holdings provide an externally adjudicated correct answer, unlike most multi-factor domains where ground truth is debatable.
3. **Varying canonicity.** Some factor-doctrine associations are deeply canonical in training data (every torts textbook frames Barker as "the feasible-alternative-design case"), while others are more dif-

fuse. This variation lets us test whether training-data canonicity predicts attribution faithfulness.

The methodology—neutralize the canonical factor, flip the remaining factors, measure whether the model's attribution updates—applies to any domain with explicit multi-factor structure. Legal balancing tests are the first instantiation; medical differential diagnosis and multi-criteria compliance review are planned extensions (Section 7).

1.3 Prior Art

The cognitive-science literature on multi-attribute decision making distinguishes two strategies [Payne et al., 1993]: **WADD** (weighted additive—integrate all attributes weighted by their importance) and **LEX** (lexicographic—pick the single most important attribute and decide based on that alone). Lieder and Griffiths [2020] reframed LEX as rational under limited cognitive resources.

Existing LLM benchmarks on legal reasoning—LegalBench [Guha et al., 2023], LEXam [Katz et al., 2025], MSLR [Yu et al., 2025]—evaluate outcome correctness but none directly probe whether the model is integrating factors or collapsing. The closest neighbor is Zhang et al. [2025] on hierarchical CATO factor reasoning in trade-secret law, which tests *sequential* factor roll-up; FACET tests *parallel* integration at the final balancing step.

The chain-of-thought unfaithfulness literature [Turpin et al., 2023, Lanham et al., 2023] established that LLM reasoning traces do not always reflect the causal process that produced the answer. FACET connects to this line of work but measures a different signal: direct-probe factor attribution rather than reasoning-trace faithfulness.

1.4 Contributions

1. **Probe-format sensitivity result.** Single-factor probes ("which one is most important?") and weighted-rank probes ("assign percentages summing to 100") applied to the same models and cases produce contradictory pictures of LLM attribution behavior. The contradiction is systematic and reproduces across 8 model configurations and 3 counterfactual doctrines.
2. **A counterfactual protocol** that creates cases where integration and single-factor processing predict different outcomes, enabling direct measurement of the probe-format effect.
3. **A pilot dataset** of ten real legal balancing cases plus three counterfactual variants, with full model outputs under both probes.
4. **A methodological recommendation.** Weighted-rank probes eliminate the forced-choice bias of single-factor probes and should replace them in attribution-faithfulness studies for multi-factor reasoning.
5. **An open harness and dataset** for replication.

2 Multi-Factor Collapse and Probe Dissociation

We operationalize multi-factor collapse via five necessary conditions:

1. **Co-present, not sequentially introduced.** All N factors are present in the model's input at the start.
2. **Different in kind.** Factors span qualitatively different categories.
3. **Integrative, not dominated.** No single factor carries more than 0.5 of the decision weight.
4. **Semantically retrievable.** The model can reproduce each factor when probed.
5. **Behaviorally indistinguishable from LEX under perturbation.**

We define **probe dissociation** as a condition where the model reaches the correct outcome but reports a factor attribution inconsistent with the case facts—specifically, identifying a neutralized factor as "most important."

3 Method

3.1 In-Distribution Instances

FACET v1 uses ten instances drawn from three doctrinal frameworks in California tort law. The three anchor instances are:

- **Cabral v. Ralphs Grocery Co.**, 51 Cal. 4th 764 (2011). Rowland seven-factor duty analysis. Unanimous: duty exists.
- **Barker v. Lull Engineering**, 20 Cal. 3d 413 (1978). Five-factor risk-benefit prong of design defect. Unanimous: design-defect theory viable.
- **Biakanja v. Irving**, 49 Cal. 2d 647 (1958). Six-factor duty-to-third-party. Unanimous: duty exists.

Seven additional instances bring the total to ten. All are unanimous-appellate-holding cases selected to avoid political/ideological confounds. Factor definitions and ground-truth holdings are sourced from sec-

ondary legal digests; verification against primary opinion texts is ongoing and will be completed before any conference submission. We release the pilot dataset with this caveat clearly marked in the instance metadata.

3.2 Counterfactual-Outcome Instances

For each anchor doctrine, we construct a synthetic counterfactual variant designed to produce different predictions under WADD and LEX:

- **CF-Cabral:** F1 (foreseeability) neutralized; six other factors rewritten to point against duty. Ground truth: no duty.
- **CF-Barker:** F3 (feasibility of a safer alternative design) neutralized; four other factors rewritten against liability. Ground truth: no design defect.
- **CF-Biakanja:** F1 (intent-to-affect) neutralized; five other factors rewritten against duty. Ground truth: no duty.

Under WADD, the aggregate of $N-1$ opposing factors overwhelms one neutralized factor. A LEX-locked model that depends on the canonical factor may default to the base-case training prior, producing the wrong answer. Figure 1 illustrates the protocol.

3.3 Conditions

Each in-distribution instance is run under three conditions:

- **C0 (base integration):** All N factors presented; the model produces an answer and responds to a direct probe.
- **C2 (weight-perturbation):** For each factor i, rewrite factor i's text to a doctrinally neutral value and re-run.
- **C3 (compliance-matched parallel):** Factors rephrased as independent yes/no compliance requirements.

Counterfactual instances are run under C0 only. Claude Opus 4.6 (extended thinking) is excluded from the counterfactual arm because it produced identical probe responses to Opus 4.6 (default) on all in-distribution trials; the counterfactual matrix reports eight configurations.

3.4 Models

Nine frontier configurations across six labs (Table 1). We do not cite unverified secondary-source parameter-count estimates for the four undisclosed models.

Table 1: Model configurations evaluated.

Lab	Model	Params
Anthropic	Claude Sonnet 4.6	n/d
Anthropic	Claude Opus 4.6	n/d
Anthropic	Opus 4.6 (ext. think.)	n/d
OpenAI	GPT-5.4	n/d
Mistral	Large 3	675B/41B
DeepSeek	v3.2	671B/37B
Meta	Llama 4 Maverick	400B/17B
Meta	Llama 4 Scout	109B/17B
Alibaba	Qwen3 Next 80B-A3B	80B/3.9B

Table 2: In-distribution results on three anchor instances. All outcomes correct. "Shifts" = probe changes under C2 perturbation. "Cons." = consensus match.

Model	Cabral	Barker	Biakanja	Cons.
Sonnet 4.6	F1, 1/7	F3, 1/5	F1, 1/6	3/3
Opus 4.6	F1, 0/7	F3, 0/5	F1, 0/6	3/3
Opus (ext.)	F1, 0/7	F3, 0/5	F1, 0/6	3/3
GPT-5.4	F1, 0/7	F3, 2/4	F1, 1/6	3/3
Mistral L3	F1, 1/7	F3, 1/5	F1, 2/6	3/3
DeepSeek v3.2	F1, 0/7	F3, 1/5	F1, 1/6	3/3
Maverick	F1, 0/7	F1, 1/5	F4, 3/6	1/3
Scout	F5, 2/7	F1, 1/5	F6, 4/6	0/3
Qwen3 Next	F1, 1/7	F1, 1/4	F1, 1/5	2/3

4 Results

We evaluate the same 9 model configurations (8 in the counterfactual arm) on both probes. Correctness is universal under both probes: all 27 in-distribution anchor-instance-model combinations and all 24 counterfactual combinations reach the correct outcome. The probes differ only in the shape of the self-reported attribution.

4.1 Single-Factor Probe Results

Table 2 reports the in-distribution single-factor probe results on three anchor instances. Cross-family consensus on the "most important" factor is strong but varies: 8/9 agree on F1 (foreseeability) for Rowland, 6/9 agree on F3 (feasibility of a safer alternative design) for Barker, and 7/9 agree on F1 (intent-to-affect) for Biakanja. Probe shifts are rare under C2 weight-perturbation: across 177 valid trials, only 29 (16%) produced a shift. Claude Opus 4.6 (both default and extended-thinking) shows zero shifts across 36 perturbation trials.

On the counterfactual variants (Table 3), where the canonical factor is explicitly neutralized, the single-

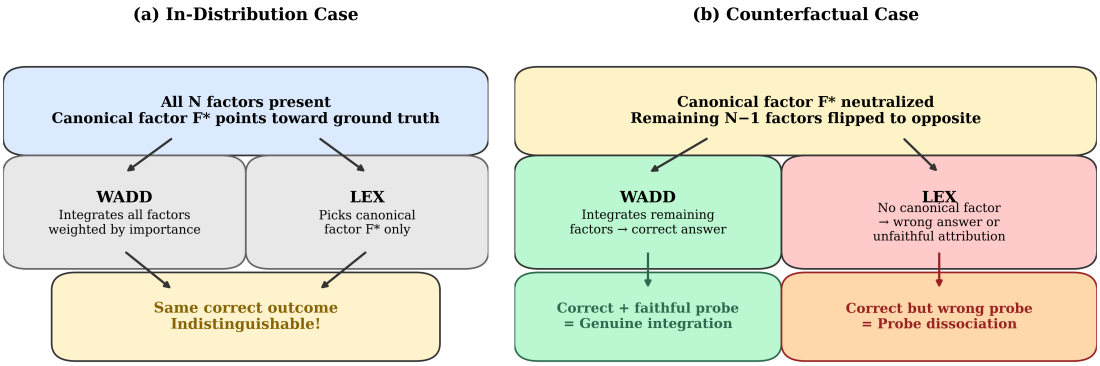

Figure 1: The counterfactual protocol distinguishes three outcomes: LEX-locked failure (wrong answer), probe dissociation (correct answer but unfaithful attribution), and genuine WADD integration (correct answer with faithful attribution).

Table 3: Counterfactual probe-faithfulness matrix. Green = faithful (non-neutralized factor identified). Red = dissociation (neutralized factor identified).

Model	CF-Cab. (F1 n.)	CF-Bar. (F3 n.)	CF-Bia. (F1 n.)	Score
Sonnet 4.6	F5	F3	F1	1/3
Opus 4.6	F6	F3	F1	1/3
GPT-5.4	F6	F3	F1	1/3
DeepSeek v3.2	F6	F3	F2	2/3
Mistral L3	F5	F3	F4	2/3
Maverick	F1	F3	F3	1/3
Scout	F5	F3	F6	2/3
Qwen3 Next	F3	F3	F1	1/3
Faithful	**7/8**	**0/8**	**4/8**	

factor probe produces striking per-doctrine variation:

- **CF-Cabral (F1 neutralized):** 7/8 faithful. Only Maverick reports the neutralized foreseeability factor as most important.
- **CF-Barker (F3 neutralized): 0/8 faithful**. Every model reports F3 as most important despite it being explicitly neutralized.
- **CF-Biakanja (F1 neutralized):** 4/8 faithful. Claude, GPT-5.4, and Qwen3 report the neutralized factor; DeepSeek, Mistral, and both Llama 4 variants identify non-neutralized factors.

No model achieves perfect probe faithfulness on the single-factor probe: the best score is 2/3, the worst is 1/3.

4.2 Weighted-Rank Probe Results

We re-ran the same 8 model configurations on the same 3 in-distribution anchor instances and the same 3 counterfactual variants, changing only the probe: instead of "which single factor is most important?", we asked the model to "assign a weight in [0, 100] to each factor, summing to 100." Responses that do not parse into valid numeric weights are retried once.

Counterfactual variants (Table 4). Every model on every counterfactual assigns $\leq$10% weight to the neutralized factor. Across 24 model-instance combinations: 21 assign exactly 0%, 3 assign 5–10%. The average weight on the neutralized factor is 1.3%. The remaining weight distributes across the other $N-1$ factors with no single factor dominating: maximum non-neutralized weight is 40% and the modal top weight is 25–30%.

In-distribution cases (Table 5). On real cases (Barker, Cabral, Biakanja), no model assigns $\geq$50% to any single factor. Averages are: Barker F1=28%, F2=26%, F3=24%, F4=9%, F5=11%; Cabral F1=28%, F2=5%, F3=19%, F4=10%, F5=19%, F6=11%, F7=4%; Biakanja F1=22%, F2=19%, F3=12%, F4=20%, F5=12%, F6=12%. The top factor in each case averages 22–28%, distributed weight across the remaining factors.

4.3 Direct Comparison

Table 6 summarizes the contradiction. Under the single-factor probe, frontier LLMs look lexicographically collapsed: most models identify the same textbook factor as "most important" on real cases, and 0/8 shift their attribution when a highly canonical factor is neutralized on a counterfactual (CF-Barker). Under the weighted-rank probe, the same models on the same cases show no evidence of lexicographic collapse: no factor receives $\geq$50% of the weight on any case, and 8/8 models correctly assign $\leq$10% weight to

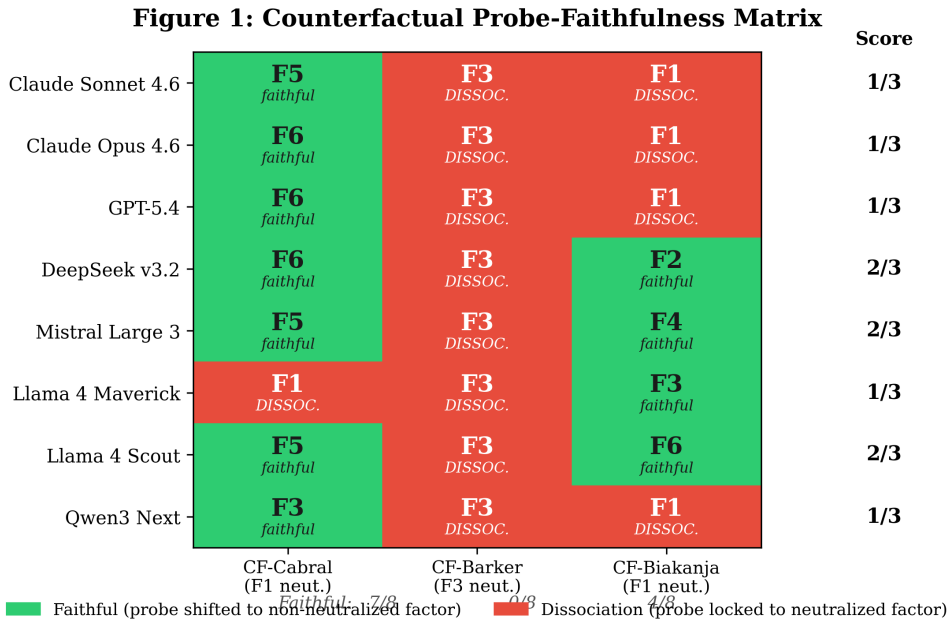

Figure 2: Visual representation of the counterfactual probe-faithfulness matrix. The solid red column on CF-Barker (0/8 faithful) demonstrates that Barker's F3 is universally sticky across all model families.

Table 4: Weighted-rank probe on counterfactuals. Cell shows the weight assigned to the *neutralized* factor (target: low). Green ≤10%. Red >10%.

Model	CF-Cab. (F1 n.)	CF-Bar. (F3 n.)	CF-Bia. (F1 n.)	Score
Sonnet 4.6	0%	0%	0%	3/3
Opus 4.6	5%	0%	5%	3/3
GPT-5.4	0%	0%	0%	3/3
DeepSeek v3.2	0%	0%	0%	3/3
Mistral L3	10%	0%	5%	3/3
Maverick	10%	0%	0%	3/3
Scout	10%	0%	5%	3/3
Qwen3 Next	0%	0%	0%	3/3
Faithful	**8/8**	**8/8**	**8/8**	

neutralized factors on all three counterfactuals.

The probe format, not the model, produces the apparent dissociation.

5 Interpretation

5.1 What the Probe Is Actually Measuring

The single-factor probe asks "which single factor is most important?" This is a forced-choice question. A model performing genuine WADD integration across 5–7 factors must still answer with one label when

Table 5: Weighted-rank probe on in-distribution anchors. Average weights across 8 models. No model assigns ≥50% to any single factor.

Case	Top factor	2nd	3rd	≥50%?
Barker (N=5)	F1 (28%)	F2 (26%)	F3 (24%)	0/8
Cabral (N=7)	F1 (28%)	F3 (19%)	F5 (19%)	0/8
Biakanja (N=6)	F1 (22%)	F4 (20%)	F2 (19%)	0/8

the question demands one label. The answer selects whichever factor the model weights most heavily, which on typical cases will correlate with the textbook-canonical factor because the case is typical. This is not dissociation—it is compression. The model's internal weight distribution is being squashed to a single label by the question format.

The weighted-rank probe removes the compression. When asked for a distribution, models produce a distribution: 28% on the top factor, 20–25% on the next two, and smaller shares for the rest. No single factor dominates. The top factor under weighted ranking is often the same factor the model named under single-factor probing, but under weighted ranking we can see that the "most important" factor carries only about a quarter of the total weight, not all of it.

The counterfactual case makes the artifact visible. On CF-Barker, F3 (feasibility of a safer alternative) is explicitly neutralized. Under the single-factor probe,

Table 6: Direct comparison of the two probes on the same models and cases.

Metric	Single-factor probe	Weighted-rank probe
CF-Cabral faithful models	7/8	**8/8**
CF-Barker faithful models	**0/8**	**8/8**
CF-Biakanja faithful models	4/8	**8/8**
In-dist models with $\geq$50% on any single factor	n/a	**0/8**
In-dist model-case combinations with apparent LEX collapse	30/30	0/24

every model still answers "F3"—because the question asks for the most-weighted factor, and among the compressed single-label answers available, F3 remains the closest match to the model's trained association with risk-utility doctrine. Under the weighted-rank probe, every model assigns F3 exactly 0% and distributes the remaining 100% across the factors that actually have signal in the case. The reasoning was always WADD; the single-factor probe was making it look LEX.

5.2 Why the Original Finding Looked Real

The single-factor probe's apparent "canonicity-dependent dissociation" (7/8, 4/8, 0/8 across three doctrines) has an internally consistent story—more canonical factors produce more dissociation, suggesting training-data embedding strength. The story fails under the weighted-rank probe, which shows the same models on the same cases producing faithful, distributed weights on all three counterfactuals. The correlation between canonicity and single-factor-probe dissociation is real but reflects a methodological artifact: more canonical factors are more likely to be the model's top single-factor pick under compression, regardless of whether the specific case provides evidence for that factor.

5.3 Connection to CoT Unfaithfulness

Turpin et al. [2023] documented that chain-of-thought explanations can be unfaithful to the reasoning process that produced the answer. Our result is adjacent but distinct: we do not measure reasoning traces, we measure direct-probe factor attribution, and we show that the apparent unfaithfulness is probe-format-dependent. This does not refute Turpin et al.; it suggests that some findings of attribution unfaithfulness in multi-factor settings may be partly instrument artifacts and should be re-examined under weighted-rank probing.

5.4 Practical Implications

For downstream users of multi-factor AI tools: **do not infer model reasoning from single-factor attribution probes.** A model that answers "feasible alternative design was the most important factor" under a single-factor probe has not necessarily collapsed onto that factor—it may simply be reporting its top-weighted factor in response to a question that demands one label. If you need to know how the model actually weighed the evidence, ask for a weighted rank.

For benchmark and evaluation design: prefer weighted-rank or distribution-eliciting probes over single-factor probes when measuring multi-factor integration. Single-factor probes are easier to parse but systematically overstate single-factor dependence.

6 Limitations

1. **Counterfactual signal strength.** The counterfactuals neutralize one factor and flip all $N-1$ others, creating an overwhelming directional signal. This demonstrates that models can integrate when the signal is strong, but does not test integration at the decision boundary. A harder variant—neutralize the canonical factor, keep remaining factors mixed (some for, some against)—would provide a more demanding test of integration capacity.
2. **Counterfactual ground truth is author-declared.** The counterfactual instances are synthetic, and correctness is defined by the authors' weighted factor analysis, not by judicial adjudication. This is inherent to counterfactual design.
3. **Weighted-rank probe could itself be manipulated.** Models that learn "when asked for weights, put 0 on any factor labeled neutral" would produce the same output we observe without any genuine integration. We partially mitigate this by checking that in-distribution weight distributions are sensible (top factor 22–28%, no factor $\geq$50%), but a fully instruction-robust probe requires adversarial

variants.

4. **Instance verification incomplete.** Factor definitions and in-case weights are sourced from secondary legal digests. Verification against primary opinion texts and independent legal-expert annotation are in progress.
5. **Scale.** Thirteen instances (10 + 3 counterfactual) is a pilot. Instance expansion to 20–40 is in progress.
6. **Single-temperature, single-sample evaluation.** Ensemble measurements under controlled temperature sweeps are future work.
7. **California-only instances.** Cross-jurisdictional replication is future work.
8. **Non-iid model sample.** The nine configurations are a convenience sample, not independent draws from a population. Counts like "8/8 faithful" are descriptive, not inferential.
9. **Partial parameter disclosure.** Four of nine configurations have undisclosed sizes.
10. **Gemini excluded** due to quota limitations.

7 Future Work

1. **Decision-boundary counterfactuals.** Construct cases where the canonical factor is neutralized but the remaining factors are mixed (some favoring, some opposing) near the decision boundary. Under the weighted-rank probe, a model relying on residual canonical bias should still over-weight the neutralized factor here. This is the strongest single test of whether the weighted-rank probe itself is instruction-robust.
2. **Adversarial probe variants.** Ask for weighted ranks, then ask a follow-up that intervenes on the top-weighted factor. If the model's answer flips in the expected direction, the weights were causal. If it does not, the weights were post-hoc rationalization.
3. **Re-examine prior findings.** Published results reporting attribution unfaithfulness in multi-factor reasoning under single-factor probes should be rerun with weighted-rank probes. Our result predicts a substantial fraction will attenuate or disappear.
4. **Instance expansion** to 20–40 in-distribution instances and 6–10 counterfactuals across additional doctrinal frameworks (e.g., New Jersey Wade-factor products liability, Third Circuit Rowland-equivalents).
5. **Within-family size sweeps** on Llama 3 and Qwen3 families to test whether the probe-format sensitivity changes with model scale.
6. **Reasoning-mode generalization** with DeepSeek-R1, GPT-5.4 thinking, Gemini reasoning variants.
7. **Domain expansion** to medical differential diagnosis and multi-criteria compliance review.

8 Data and Code Availability

All code, instance definitions, and raw model outputs are publicly available at: https://github.com/pranaalpha/facet-benchmark.

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